

MDHGNN Project Description

Weekly Frequency Prediction for Copper Futures Based on Yield Modeling

I. Project Overview

This project implements the Multi-channel Dual-path Hypergraph Neural Network (MDHGNN) for weekly price forecasting of copper futures. The model's primary prediction objective is not the price itself but the future yield vector; price forecasts are derived from the previous copper price and predicted yields, while directional accuracy is determined by the sign of the predicted yields.

The core concept of this project is to learn two types of information simultaneously:

- Temporal Path: Uses GRUs and attention pooling to learn the temporal dependencies of copper prices and related market variables.
- Structure path: Treat input features as hypergraph nodes and learn group relationships and higher-order relationships between variables through multi-channel hyperedges.

II. Data and Sample Period

To address the issue of copper data ending in 2016 in the previous version, this update has revised the copper futures data while retaining the old copper price files as backups.

project	explain
Copper Futures Data Source	Yahoo Finance chart data, ticker 'HG=F'
Daily Price Range for Copper Futures	From January 3, 2006 to March 27,2026, a total of 5,092 rows
Weekly Copper Data	After resampling, there are a total of 1,056 rows.
external variable	WTI crude oil, gold, silver, and the US dollar index
Finally align the frequency samples	From July 22,2011 to March 27,2026, a total of 767 rows
Feature Count	30 items, including copper futures OHLCV, technical indicators, and external market variables
Frequency Window Length	'W = 12'

Since silver futures data became available starting July 22,2011, the final multivariate modeling sample also begins on that date. The original copper price records span 2006–2026, but observations prior to the common external variable period are excluded from the final multivariate analysis.

data partitioning

The training set, validation set, and test set are divided in chronological order without random shuffling.

Prediction Step Size	Training Target Sample	Verify Target Sample	Test Target Sample
h = 1	524	77	153
h = 2	524	77	152

Prevent data leakage

1. Training, validation, and testing are divided in chronological order; no random division across periods is performed.
2. Z-score normalization uses only the mean and standard deviation of the training set, then applies them to the validation and test sets.
3. The input window for sample t contains only historical data and does not include target-period prices.
4. External variables are first aligned by date and resampled to a weekly frequency before proceeding to the feature engineering process.
5. Missing values are handled with forward filling; backward filling is not used.

III. Prediction Objectives

Input window:

$$X_t = [x_{t-W}, \dots, x_{t-1}]$$

The return at step j is defined as:

$$r_{t,j} = (P_{t+j-1} - P_{t-1}) / P_{t-1}, j = 1, \dots, h$$

The price reduction formula is:

$$\hat{P}_{t+j-1} = P_{t-1} * (1 + r_{t,j})$$

Thus, when $h=1$, the objective is the return rate of the following week relative to the previous week; when $h=2$, the second-stage objective is the two-week cumulative return rate calculated from the same baseline price P_{t-1} .

IV. Model Structure

MDHGNN includes two paths:

module	explain
Time Path	Learn temporal dependencies from the input window, with the core components being GRUs and attention pooling.
Structure Path	Use 30 input features as nodes and employ hyperedges to represent higher-order relationships between variables.
Integration Layer	Use a globally learnable weight fusion of temporal and structural representations.
Prediction Head	Output the future yield at step h .

The hyperedge types are as follows:

type	Construction Method	quantity
Correlation Soft Super Edge	Generate soft associations based on feature similarity in the input window.	90
Domain Prior Hyper-edge	It is established based on economic indicators such as price, trend,	9

	momentum, volatility, trading volume, external markets, and overall market conditions.	
Adaptive Hyperedge	It is generated through learnable projections based on feature node embeddings.	8
amount to	The node has 30 input features.	107

V. Baseline Methods and Metrics

method	explain
Naive persistence	With a parameter-free persistent baseline, the predicted return is always 0, indicating that the price in the next period equals that of the previous period.
GRU	Two-layer GRU regression model.
MLP	After expanding the input window, use a multi-layer fully connected network for regression.
Transformer	Transformer Encoder Regression Baseline.

The evaluation metrics include price-related indicators such as RMSE, MAE, MAPE, and R^2 ; return-related metrics like Ret RMSE and Ret R^2 ; directional indicators including DA, DA up, and DA down; and strategy-based metrics such as Sharpe ratio, Annually Return, and Max Drawdown.

The Ret RMSE is first calculated in decimal yield units, then multiplied by 10^4 to convert it into basis points:

$$\text{Ret_RMSE}(\text{bp}) = 10^4 * \sqrt{(1/N) * \sum((r_i - \hat{r}_i)^2)}$$

VI. Main Experimental Results

The following results were regenerated by the current code on a shared real sample set at weekly frequency using random seeds 42,123, and 456. The values in the table represent mean $\pm$ standard deviation.

h = 1 Result:

method	RMSE	MAE	MAPE (%)	R2	DA (%)
MDHGNN	0.1908 $\pm$ 0.0036	0.1240 $\pm$ 0.0041	2.76 $\pm$ 0.10	0.9091 $\pm$ 0.0035	48.37 $\pm$ 2.33
GRU	0.1945 $\pm$ 0.0032	0.1315 $\pm$ 0.0040	2.91 $\pm$ 0.08	0.9056 $\pm$ 0.0031	50.33 $\pm$ 0.92
MLP	0.1953 $\pm$ 0.0030	0.1303 $\pm$ 0.0027	2.87 $\pm$ 0.06	0.9048 $\pm$ 0.0029	46.62 $\pm$ 1.72
Transformer	0.1997 $\pm$ 0.0155	0.1312 $\pm$ 0.0150	2.92 $\pm$ 0.35	0.8999 $\pm$ 0.0159	52.72 $\pm$ 1.63
Naive	0.1876 $\pm$ 0.0000	0.1200 $\pm$ 0.0000	2.66 $\pm$ 0.00	0.9122 $\pm$ 0.0000	47.06 $\pm$ 0.00

At $h=1$, MDHGNN's price error performance surpasses that of the tested neural network baseline but does not exceed that of Naive persistence.

h = 2 Result

method	RMSE	MAE	MAPE (%)	R2	DA (%)
MDHGNN	0.2781 ± 0.0241	0.1952 ± 0.0231	4.36 ± 0.51	0.8062 ± 0.0342	55.04 ± 1.35
GRU	0.2643 ± 0.0178	0.1868 ± 0.0189	4.15 ± 0.42	0.8255 ± 0.0239	52.08 ± 1.38
MLP	0.2592 ± 0.0020	0.1793 ± 0.0009	3.99 ± 0.01	0.8328 ± 0.0025	53.29 ± 2.56
Transformer	0.2593 ± 0.0087	0.1757 ± 0.0094	3.93 ± 0.24	0.8326 ± 0.0112	55.92 ± 1.63
Naive	0.2512 ± 0.0000	0.1698 ± 0.0000	3.76 ± 0.00	0.8431 ± 0.0000	45.07 ± 0.00

At $h=2$, the advantage of MDHGNN does not persist; Naive, Transformer, and MLP demonstrate superior performance in terms of price error.

VII. Ablation Experiment

The ablation experiment employed a masked ablation protocol: each random seed first trained the complete MDHGNN model, then during inference at the same weight settings, one hyper-edge channel or temporal path was masked.

variant	RMSE	R2	explain
Full MDHGNN	0.1908 ± 0.0036	0.9091 ± 0.0035	Complete the reasoning chain.
w/o Corr-HE	0.2004 ± 0.0092	0.8996 ± 0.0090	Shrink the correlation soft superedge.
w/o Domain-HE	0.2003 ± 0.0091	0.8997 ± 0.0089	Block prior hyperedges in the field.
w/o Adapt-HE	0.2003 ± 0.0091	0.8997 ± 0.0089	Hide adaptive hyperedges.
w/o GRU	0.2821 ± 0.1119	0.7702 ± 0.1795	Hide the GRU timing path.

The results indicate that the GRU temporal path is particularly critical for the current task; there are minimal differences among the three types of hyper-edge channels, and it would be inaccurate to claim that each hyper-edge channel independently contributes significantly.

VIII. Operating Method

Installation dependencies:

```
pip install -r requirements.txt
```

Main Comparative Experiment:

```
python experiments/run_all.py --frequency weekly --horizons 1 2 --seeds 42 123 456
```

Masked Ablation Experiment:

```
python experiments/run_masked_ablation.py --frequency weekly --horizon 1 --seeds 42 123 456
```

Single Model Training:

```
python main.py --frequency weekly --horizon 1
```

IX. Project Structure

File/Directory	explain
`config.py`	Centralize management of data, models, and training parameters.
`main.py`	Single-model training entry.
`data/`	Data reading, resampling, feature engineering, preprocessing, and datasets.
`models/`	Implementation of MDHGNN, supergraph convolution, prediction head, and baseline methods.
`experiments/run_all.py`	Conduct a main comparison experiment and generate result tables, DM tests, and charts.
`experiments/run_masked_ablation.py`	Masked ablation experiment.
`experiments/redraw_publication_figures.py`	Redraw the publication at the checkpoint.
`results/tables/`	JSON metrics, DM validation, ablation summaries, and experiment manifest.
`results/figures/`	PNG, PDF, SVG charts.
`tests/`	unit test .
`CODE_CHANGES.md`	This document explains the modifications to code and data compared to the previous version.

X. Version Description

The key changes from the previous code version include updating copper futures data to March 27,2026 and synchronizing the experimental sample period from January 3, 2006 to March 27,2026. The core model structure remains unchanged, and the experiment was rerun using the updated shared real sample period.